

sedphot

User Manual

Galaxy in, SED photometry out

Version 0.3.0

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1 What this does

sedphot turns a galaxy into an SED-ready photometry table.

You give it a galaxy name (or a position on the sky) and an output directory. It then does some or all of the following, depending on which command you run:

- Looks the galaxy up in the public catalog archives (GALEX, SDSS, J-PLUS, Pan-STARRS, Legacy Surveys, AllWISE, HST) and records the published flux of the nearest source in every band.
- Downloads survey images (SDSS, Pan-STARRS, Legacy Surveys, CFHT, HST) and measures the galaxy itself in every band through one identical procedure, so the colors are internally consistent.
- Optionally requests SPHEREx spectrophotometry from IRSA.
- Writes a combined SED figure, quality-assurance figures, and a JSON record of exactly how every number was produced.

Fluxes come out in microjansky on the AB system. Everything for one galaxy lands under one directory that you choose.

2 Installing

These instructions assume the machine has nothing set up yet.

2.1 Get Python

sedphot needs **Python 3.11 or newer** and a stack of scientific libraries (NumPy, SciPy, Astropy, Astroquery, Matplotlib, and others).

Use **Miniconda**. Conda installs Python and the scientific libraries as pre-built binaries, which avoids the compiler toolchain that some of these packages otherwise require, and it keeps each project's libraries in a separate environment so one project cannot break another.

Windows

1. Go to <https://www.anaconda.com/download/success> (or <https://docs.conda.io/projects/miniconda/>) and download the **Miniconda Windows 64-bit installer (.exe)**.
2. Run the installer. Accept the defaults.
3. From the Start menu, open **Anaconda Prompt**. Every command in this manual is typed there.

macOS

1. Download the **Miniconda macOS installer** for your chip (Apple silicon = **arm64**, Intel = **x86_64**) from the same page.
2. Run the installer, then open **Terminal**.

Linux

1. Download the Miniconda Linux **x86_64** installer script from the same page.
2. Run it with **bash Miniconda3-latest-Linux-x86_64.sh**, then open a new terminal.

Check that conda is available:

```
conda --version
```

You should see something like **conda 24.9.2**.

2.2 Get the code

The source lives at <https://github.com/chrishopp12/query-photometry>.

With git. On Windows, install Git for Windows from <https://git-scm.com/download/win> first; macOS and Linux usually have git already. Then:

```
git clone https://github.com/chrishopp12/query-photometry.git
```

This creates a folder named `query-photometry` in the current directory. Move into it:

```
cd query-photometry
```

Without git. Open <https://github.com/chrishopp12/query-photometry> in a browser, click the green **Code** button, choose **Download ZIP**, and unzip it. You get a folder named `query-photometry-main`. Move into it:

```
cd query-photometry-main
```

You are in the right folder if it contains a file named `pyproject.toml`.

2.3 Create an environment and install

Conda route (recommended, all platforms).

```
conda create -n sedphot python=3.12
```

Answer `y` when it asks. Then activate it:

```
conda activate sedphot
```

Your prompt now starts with `(sedphot)`. Install the package:

```
pip install -e .
```

That reads `pyproject.toml` and pulls in every dependency.

If any dependency fails to build (rare, but `reproject` is the usual suspect), install the scientific stack from conda-forge first and then install `sedphot` without re-resolving them:

```
conda install -c conda-forge numpy scipy pandas astropy
```

```
conda install -c conda-forge astroquery matplotlib pillow
```

```
conda install -c conda-forge reproject requests defusedxml
```

```
pip install -e . --no-deps
```

pip / venv route (no conda). Only use this if Python 3.11+ is already installed and on your `PATH`.

Windows (Anaconda Prompt, PowerShell, or Command Prompt):

```
python -m venv .venv
```

```
.venv\Scripts\activate
```

macOS / Linux:

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

Then, on any platform:

```
pip install -e .
```

2.4 What “editable” means

`pip install -e .` installs the package **in place**: Python runs the source files in the folder you cloned, rather than a copy hidden away in the environment. If you `git pull` a newer version, the change takes effect immediately with no reinstall.

2.5 Verify the install

Three checks, in increasing strength.

Is the command there?

```
sedphot --help
```

Correct output begins:

```
usage: sedphot [-h]
               {resolve,catalogs,measure,spherex,spherex-plan,
               spherex-batch,sed,plan,overlay,remeasure,run,batch}
               ...
```

Galaxy in, SED photometry out: multi-archive retrieval and measurement.

Is the right version installed?

```
python -c "import sedphot; print(sedphot.__version__)"
```

Correct output:

```
0.3.0
```

Can it reach the network? This resolves a real galaxy name against the CDS Sesame service:

```
sedphot resolve --name "NGC 4889"
```

Correct output:

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)
RA   = 195.03373750
Dec  = +27.97702500
```

```
label = NGC_4889
```

If all three work, the install is good.

Every later session needs the environment activated first:

```
conda activate sedphot
```

3 Before you start

The mental model is small.

You pick a galaxy. Either by name (`--name "NGC 4889"`, resolved through Sesame, then NED, then SIMBAD) or by explicit position (`--ra 195.033738 --dec 27.977025`). Give one or the other, never both.

You pick an output directory. `--out-dir` is required on every command that writes anything. Everything for that galaxy lands under it, inside a subdirectory called **Photometry/**: the tables, the figures, the downloaded images, and the JSON records. One galaxy, one directory.

Files are named after a label. The label defaults to the sanitized name (NGC 4889 becomes NGC_4889) or, for a bare position, an IAU-style designation (J130008.10+275837.3). Override it with `--label`. Output files are `<label>_catalog.csv`, `<label>_measured.csv`, `<label>_sed.png`, and so on.

Downloads are cached. The first run of a galaxy downloads images and catalog queries; later runs on the same directory reuse them. Re-running a command is cheap.

That is the whole model. A typical directory choice looks like:

```
sedphot run --name "NGC 4889" --out-dir NGC_4889
```

or, if you keep galaxies together:

Windows

```
sedphot run --name "NGC 4889" --out-dir galaxies\NGC_4889
```

macOS / Linux

```
sedphot run --name "NGC 4889" --out-dir galaxies/NGC_4889
```

Path separators are the only difference between platforms in this manual. Everything else is identical.

4 The verbs

sedphot has twelve commands. The first eight work on one galaxy; the last four work on a list of them.

Verb	What it does	What it writes
resolve	Turns a name into coordinates and an output label.	Nothing. Prints only.
catalogs	Asks each catalog archive for its published photometry of the nearest source.	<code><label>_catalog.csv</code> + sidecar, <code>coverage_catalogs.json</code>

Verb	What it does	What it writes
measure	Downloads survey images and measures the galaxy in every band.	<label>_measured.csv + sidecar, coverage_measure.json, cached images, per-band QA figures, QA/growth_curves.png
spherex	Requests a SPHEREx spectrophotometry table from IRSA.	SPHEREx/table_photometry.<tag>.csv + sidecar, SPHEREx/extractions.json
sed	Draws the combined SED from tables already on disk. With --spherex , draws the SPHEREx spectrum under the broadbands as a second, separate figure.	<label>_sed.png, or <label>_spherex_sed.png under --spherex
overlay	Draws every archive's matched position on the HST color image.	<label>_overlay.png, cached HST composite
remeasure	Re-reports fluxes from a finished measurement at a different aperture.	Nothing, unless you pass --out (a CSV) or --write-qa (figures).
run	catalogs , then measure , then optionally spherex , then sed .	Everything the stages above write, except the SPHEREx figure.

And four for a list of galaxies:

Verb	What it does	What it writes
plan	Reads a list of targets and works out the order they can safely be measured in.	<plan>.json and a .csv of the same rows, for you to read
batch	Measures every target in a plan, several at once.	Everything run writes, for each target, plus registry files, a report and per-target logs
spherex-plan	Groups targets that sit close enough together on the sky to contaminate one another into joint SPHEREx jobs.	<config>.json and a .csv of every member, for you to review
spherex-batch	Runs the joint SPHEREx jobs in a config.	One SPHEREx table per galaxy, exactly as spherex writes, plus a report and per-group logs

--out-dir is **required** on **catalogs**, **measure**, **spherex**, **sed**, **overlay**, and **run** — the six verbs that write into one galaxy's directory. **resolve** takes no output directory because it writes nothing. **remeasure** takes a path to a finished measurement's JSON record instead. The four list verbs write in several places at once, so each output is named separately: **--out** for a plan or config, **--report** for a report, **--registry-dir** for **batch**'s registries, **--log-dir** for logs.

5 Your first run

This walks one galaxy from nothing to a measured table, using only the Legacy Surveys images so it finishes in a couple of minutes.

Step 1. Activate the environment.

```
conda activate sedphot
```

Step 2. Confirm the galaxy resolves.

```
sedphot resolve --name "NGC 4889"
```

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)
RA   = 195.03373750
Dec  = +27.97702500
label = NGC_4889
```

Step 3. Fetch the catalog photometry.

```
sedphot catalogs --name "NGC 4889" --all --out-dir NGC_4889
```

The console shows one block per archive, then a summary:

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)

Target: RA=195.033738, Dec=+27.977025 (search radius=2.0")

=== galex ===
[GALEX GUVcat] Attempt 1/5, radius=2.0"
[GALEX GUVcat] Found 1 match(es) at radius=2.0"

=== sdss ===
[SDSS] Attempt 1/5, radius=2.0"
[SDSS] Found 1 match(es) at radius=2.0"

...

Provider summary:
galex      ok          2 rows
sdss       ok          5 rows
jplus     no_match     0 rows -- no J-PLUS DR3 source found
panstarrs ok          5 rows
legacy     ok          7 rows
allwise    ok          4 rows
hst        ok          2 rows

Saved 25 photometric points to: NGC_4889/Photometry/NGC_4889_catalog.csv
```

followed by the table itself.

Step 4. Measure the images.

```
sedphot measure --name "NGC 4889" --instruments legacy --out-dir NGC_4889
```

```
[resolve] 'NGC 4889' -> RA=195.033738, Dec=+27.977025 (Sesame)

Target: RA=195.033738, Dec=+27.977025 (aperture 12", scene engine)

=== legacy images ===
  3 band image(s) ready

(scene catalog and star queries report here on a first run)

[Legacy_r] seat solve [target:sersic]: nfev 168, 4.1s last, ...
[Legacy_r] background track: -0.0074 -> -0.0071 uJy/as2 (6/121 ...
[Legacy_r] fill: 4% of aperture masked, 100% twin-filled, ...
[Legacy_r] f_ap = 30184.2 +/- 41.6 uJy (empap), excess -22.4, ...
Legacy r: 30184.2 +/- 41.6 uJy (empap; QA Legacy_r.png)
[Legacy_z] ...
Legacy z: 48120.7 +/- 63.9 uJy (empap; QA Legacy_z.png)
[Legacy_g] ...
Legacy g: 12507.5 +/- 27.3 uJy (empap; QA Legacy_g.png)

Provider summary:
  legacy      ok          3 rows -- measured bands: r, z, g

Saved 3 measured bands to: NGC_4889/Photometry/NGC_4889_measured.csv
```

followed by the table. The numbers depend on the galaxy; the structure does not. The indented [Legacy_r] lines are the engine's running commentary and are truncated above to fit the page; on screen they run to the end of the line.

Step 5. Draw the SED.

```
sedphot sed --out-dir NGC_4889
```

```
[sed] wrote NGC_4889/Photometry/NGC_4889_sed.png
```

The resulting file tree.

```
NGC_4889/
+-- Photometry/
  +-- NGC_4889_catalog.csv
  +-- NGC_4889_catalog.provenance.json
  +-- NGC_4889_measured.csv
  +-- NGC_4889_measured.provenance.json
  +-- NGC_4889_sed.png
  +-- coverage_catalogs.json
  +-- coverage_measure.json
  +-- Legacy/
    | +-- legacy_ls-dr9_g.fits
    | +-- legacy_ls-dr9_r.fits
```

```
|   +-- legacy_ls-dr9_z.fits
|   +-- QA/
|       +-- Legacy_g.png
|       +-- Legacy_r.png
|       +-- Legacy_z.png
+-- QA/
|   +-- growth_curves.png
+-- scene/
    +-- gaia_scene.csv
    +-- tractor_scene_dr9.csv
```

Steps 3, 4 and 5 together are exactly what the `run` verb does:

```
sedphot run --name "NGC 4889" --out-dir NGC_4889
```

6 Common tasks

Each entry gives when to use it and the exact command. Substitute your own galaxy name and output directory.

6.1 A note on long commands

A few commands below are too wide for one line and are shown broken across two, with a trailing `\`. That is the continuation character for `bash` and `zsh` (macOS, Linux).

Shell	Continuation character
macOS / Linux (bash, zsh)	<code>\</code>
Windows Command Prompt / Anaconda Prompt	<code>^</code>
Windows PowerShell	<code>`</code> (backtick)

On Windows, either swap the trailing `\` for the right character, or — simplest and always correct — delete the line break and paste the whole thing as one long line.

6.2 Look up a galaxy's coordinates from its name

Use this before anything else, to confirm the name resolves and to see what label the output files will use.

```
sedphot resolve --name "NGC 4889"
```

If the name does not resolve, look the position up yourself (NED or SIMBAD in a browser) and use `--ra` / `--dec` everywhere instead:

```
sedphot resolve --ra 195.033738 --dec 27.977025
```

6.3 Fetch catalog photometry from every archive

The normal starting point. Queries all seven catalog archives and writes one combined table.

```
sedphot catalogs --name "M87" --all --out-dir M87
```

Each archive starts with a 2-arcsecond search radius and doubles it up to five times if nothing is found. Raise the starting radius for a galaxy whose catalog position is offset:

```
sedphot catalogs --name "M87" --all --radius 5 --out-dir M87
```

6.4 Fetch catalog photometry from one archive

Use this when you only need one archive, or when one archive was down during an earlier run and you want to retry just that one. The valid names are `galex`, `sdss`, `jplus`, `panstarrs`, `legacy`, `allwise`, `hst`.

```
sedphot catalogs --name "M87" --instruments legacy --out-dir M87
```

Several at once:

```
sedphot catalogs --name "M87" --instruments legacy sdss --out-dir M87
```

Note that the table is rewritten from scratch each time, so a run with one archive replaces a table that held all seven. Re-run with `--all` to rebuild the full table.

6.5 Measure images from every survey

Downloads images from all five image archives and measures the galaxy in every band. This is the slow command: expect minutes per band on a first run, less afterwards because the images are cached.

```
sedphot measure --name "M87" --all --out-dir M87
```

6.6 Measure images from specific surveys only

The valid names are `sdss`, `panstarrs`, `legacy`, `cfht`, `hst`.

```
sedphot measure --name "M87" --instruments legacy cfht --out-dir M87
```

6.7 Measure a subset of filters

`--bands` applies the same filter list to every selected survey; a survey that does not have one of them simply skips it.

```
sedphot measure --name "M87" --all --bands g r z --out-dir M87
```

The default filter sets are:

Survey	Default filters
SDSS	ugriz
Pan-STARRS	grizy
Legacy Surveys (dr9)	grz
Legacy Surveys (dr10)	griz
CFHT MegaCam	ugriz
HST	every filter in the matched observation group

For HST, filter names are flight-filter designations:

```
sedphot measure --name "M87" --instruments hst \
  --bands F475W F814W --out-dir M87
```

6.8 Change the aperture size

The measurement aperture is a circle centered on the target. The default radius is **12 arcseconds**.

```
sedphot measure --name "M87" --all --aperture 15 --out-dir M87
```

The aperture must fall inside the curve-of-growth radius grid, which by default runs from 2 to 40 arcseconds. To go outside that, move the grid too:

```
sedphot measure --name "M87" --all --aperture 50 \
  --radii 2 10 20 30 40 50 60 --cutout-size 200 --out-dir M87
```

6.9 Re-report an existing measurement at a different aperture

This is the fast path. Every measurement stores the *curve of growth* — the enclosed flux at every radius from 2 to 40 arcseconds — so changing the reported aperture is a lookup, not a re-measurement. It takes milliseconds and never touches the files on disk unless you ask it to.

Point it at the measurement's JSON record:

Windows

```
sedphot remeasure M87\Photometry\M87_measured.provenance.json ^
  --mode aperture --aperture 10
```

macOS / Linux

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
  --mode aperture --aperture 10
```

The result prints as a table. To save it, add `--out`:

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
  --mode aperture --aperture 10 --out M87_R10.csv
```

Two other useful forms. The total flux of the fitted model:

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
  --integrated
```

And the fitted model's flux inside a chosen radius. `--mode` defaults to `aperture`, matching `measure`, so the model flux must be asked for explicitly:

```
sedphot remeasure M87/Photometry/M87_measured.provenance.json \
  --mode sersic --aperture 20
```

Asking for an aperture **past** the stored grid in `--mode aperture` is different: there is no stored value to look up, so the command rebuilds the scene from the saved fit and integrates. It says so, and it takes minutes rather than milliseconds.

6.10 Measure a small or compact target

The defaults are sized for survey-resolution images of ordinary galaxies. A compact target — typically on an HST mosaic — needs four things moved together, or the measurement will treat the galaxy itself as sky.

Flag	Default	Compact-target value
<code>--aperture</code>	12 arcsec	2 arcsec
<code>--cutout-size</code>	120 arcsec	20 arcsec
<code>--sky-rmin</code>	15 arcsec	4 arcsec
<code>--radii</code>	2 to 40 arcsec	0.5 to 8 arcsec

`--sky-rmin` is the boundary between “target” and “sky”. Sky is measured only outside it, and no pixel inside it is allowed to vote on the background level. At the 15-arcsecond default, a compact HST target would have most of the visit excluded from the sky estimate.

```
sedphot measure --name "M87" --instruments hst --aperture 2 \
  --cutout-size 20 --sky-rmin 4 --radii 0.5 1 1.5 2 3 4 6 8 \
  --out-dir M87
```

The aperture must lie within the `--radii` grid; the grid’s outer radius must be under half the cutout size.

6.11 Run everything for one galaxy in a single command

Catalogs, then images and measurement, then the SED plot. Every stage is isolated: if one fails, the failure is recorded and the rest still run, and the command exits with a nonzero status.

```
sedphot run --name "NGC 4889" --out-dir NGC_4889
```

Add SPHEREx:

```
sedphot run --name "NGC 4889" --spherex sersic --out-dir NGC_4889
```

Leave out slow or irrelevant archives (`--skip` accepts both catalog and image archive names):

```
sedphot run --name "NGC 4889" --skip hst jplus --out-dir NGC_4889
```

`run` accepts every measurement flag `measure` does — `--aperture`, `--bands`, `--cutout-size`, `--sky-rmin`, `--radii`, and the rest — with one exception: `--dump-arrays` is available only on `measure`.

6.12 Make the SED plot from tables already on disk

Use this after editing a table by hand, or after re-running one stage. It reads whichever of `<label>_catalog.csv` and `<label>_measured.csv` exist and draws them on one axis.

```
sedphot sed --out-dir NGC_4889
```

If the directory holds tables for more than one label, name the one you want:

```
sedphot sed --label NGC_4889 --out-dir NGC_4889
```

6.13 Check that the archives all matched the same object

This is the check the flux columns cannot make: two archives confidently reporting photometry for two different objects. It draws each archive's matched position on the HST color image of the field, so a mismatch is visible.

```
sedphot overlay --out-dir NGC_4889
```

It only works where HST has imaged the field. The first run downloads a ~**380 MB** file (the HST detection mosaic, needed only for its coordinate grid) and caches it. If you already have a FITS file on the same grid, point at it and skip the download:

```
sedphot overlay --out-dir NGC_4889 \
  --wcs-from NGC_4889/Photometry/HST/hst_F814W_sci.fits
```

The file must be on the same pixel grid as the color composite; the dimensions are checked, and a mismatch refuses rather than misplacing every marker.

Adjust the panel sizes (both are half-widths in arcseconds):

```
sedphot overlay --zoom-size 3 --context-size 30 --out-dir NGC_4889
```

6.14 Fetch SPHEREx spectrophotometry

Submits a forced-photometry job to the IRSA SPHEREx tool and writes the raw per-visit table. The job runs on IRSA's servers; the command polls until it finishes. A job typically takes twenty minutes to an hour, and the command waits up to three hours for one that is running. Waiting to *start* is budgeted separately and generously (six hours), because the service runs two jobs at a time and queues the rest — a job sitting in the queue has not begun, and abandoning it would throw away work that would have completed.

The default source model is an elliptical Sérsic profile whose shape is looked up in the Legacy Surveys Tractor catalog:

```
sedphot spherex --name "NGC 4889" --out-dir NGC_4889
```

If the shape lookup fails or returns nothing usable, the command **stops** rather than guessing. Three ways forward:

Give the shape yourself, as *n*, axis ratio (*a/b*, at least 1), position angle in degrees east of north, and effective radius in arcseconds:

```
sedphot spherex --name "NGC 4889" --out-dir NGC_4889 \
  --sersic-params 4 1.3 65 12
```

Fit the shape on an image instead:

```
sedphot spherex --name "NGC 4889" --sersic-from z --out-dir NGC_4889
```

Or treat the target as a point source (this biases an extended source, in a wavelength-dependent way):

```
sedphot spherex --name "NGC 4889" --model psf --out-dir NGC_4889
```

Each distinct configuration gets its own tagged output file, so a PSF run and a Sérsic run coexist. Nothing is ever overwritten.

6.15 Fetch SPHEREx for galaxies with close neighbors

A SPHEREx pixel is about 6.2 arcseconds across — far coarser than the optical images. A neighbor a few pixels away therefore has its light inside your target’s measurement, and asking for the target on its own gives it *all* of that light.

The IRSA tool can fit several positions at once and divide the light between them. Two commands drive it: one to decide the groups, one to run them.

```
sedphot spherox-plan --targets targets.csv --out groups.json \
  --blend-radius 45
```

This reads the same target CSV that `plan` reads (its format is shown in section 6.17), groups targets within 45 arcseconds of one another, looks up each one’s shape, and adds nearby catalog neighbors that are bright enough to matter. It works offline if the galaxies have been measured already, because the shapes come from the scene catalogs already cached in each galaxy’s directory.

Read `groups.csv` before running anything. It is written for exactly that purpose — one row per source, with the group it belongs to, its position, its shape, and where that shape came from. Two things to check:

- **The role column.** `science` means this galaxy gets its own SPHEREx table, in its own directory, exactly as if you had run `spherox` on it alone. `ancillary` means the source is only there to soak up light that would otherwise be counted as your target’s; its spectrum is filed with the group, not with a galaxy.
- **Blank shape columns.** A galaxy with no shape found is listed in the config’s `planning_notes`. Fill the shape in by hand, or plan with `--model psf`.

Edit the file freely — add a member, remove one, correct a shape. Then:

```
sedphot spherox-batch --config groups.json \
  --report groups_report.csv --log-dir logs
```

Each galaxy ends up with one SPHEREx table in its own directory, named with a `joint-` tag so it sits alongside any single-position table rather than replacing it. Nothing downstream needs to know a joint fit happened.

Twenty sources per job — and no warning past that

The IRSA tool takes the first twenty rows of an uploaded list and **silently discards the rest**. `sedphot` refuses a larger group rather than letting that happen, and says which neighbors it dropped when planning hits the limit. It also checks every shape against the tool’s stated limits before submitting, because the tool does not check uploaded shapes at all — it will happily fit an impossible galaxy and hand you a table that looks perfectly normal.

If you interrupt a sweep, just run it again: a group whose galaxies already carry that exact extraction is skipped. “That exact extraction” includes the membership, so changing a group — adding a neighbor, fixing a shape — correctly re-runs it rather than reusing the old answer. The two are different measurements, and both are kept.

6.16 Process a list of galaxies one after another

Put one galaxy name per line in a plain text file, `galaxies.txt`:

```
NGC 4889
NGC 4874
M87
```

Windows — Command Prompt or Anaconda Prompt batch file. Save as `run_all.bat` next to `galaxies.txt`, then double-click it or type `run_all.bat`:

```
@echo off
for /f "usebackq delims=" %%G in ("galaxies.txt") do (
    echo === %%G ===
    sedphot run --name "%%G" --out-dir "galaxies\%%G"
)
```

Windows — PowerShell. Type this at a PowerShell prompt, or save it as `run_all.ps1`:

```
Get-Content galaxies.txt | ForEach-Object {
    Write-Host "=== $_ ==="
    sedphot run --name "$_" --out-dir "galaxies\$($_ -replace ' ','_')"
}
```

macOS / Linux — bash or zsh. Save as `run_all.sh`, make it executable with `chmod +x run_all.sh`, then run `./run_all.sh`:

```
#!/usr/bin/env bash
while IFS= read -r name; do
    [ -z "$name" ] && continue
    echo "=== $name ==="
    sedphot run --name "$name" --out-dir "galaxies/${name// /_}"
done < galaxies.txt
```

Careful

Run these one galaxy at a time, not in parallel. If you are also using `--registry-update` (see section 11.2), running in parallel will corrupt the shared file. `batch` (section 6.17) is the supported way to measure several galaxies at once without that risk.

These loops are fine for a handful of galaxies. For a real sample, use `plan` and `batch` instead — they measure several galaxies at once *safely*, which a loop cannot.

6.17 Measure a whole sample safely

The obstacle to running galaxies in parallel is the registry: it is a single shared file, and two galaxies writing it at the same time lose each other's entries. Running everything serially avoids that, but can turn a one-hour job into a day.

`plan` and `batch` solve it properly. Start with a CSV of your targets — a name column, `ra_deg` and `dec_deg`, and optionally `label` and `dir`:

```
name,ra_deg,dec_deg,dir
NGC_4889,195.033738,27.977025,galaxies/NGC_4889
NGC_4874,194.898792,27.959528,galaxies/NGC_4874
```

Work out the order:

```
sedphot plan --targets targets.csv --out plan.json
```

This looks at every field, decides which galaxies produce a shape that some *other* galaxy will read, and splits the sample in two. It writes `plan.json` and a `plan.csv` you can open in a spreadsheet.

Then run it:

```
sedphot batch --plan plan.json --registry-dir registries \
  --report report.csv --log-dir logs \
  --workers 3 --images legacy cfht sdss panstarrs
```

`batch` runs in two passes. The first handles only the galaxies that share shapes, in small groups, and gives *each group its own registry file* — so two galaxies never write the same file, and the problem the warning above describes cannot occur. It then merges those files into one and freezes it. The second pass runs everything else at once against the frozen copy, writing nothing shared.

`report.csv` gets one row per galaxy, rewritten as each finishes, and `logs/<label>.log` holds that galaxy’s output line by line while it runs.

Check the band count rather than the report

If a single band fails — a dropped download, a slow disk — the run logs it and carries on, and the galaxy still reports `ok`. The only way to notice is to count the rows in each galaxy’s `_measured.csv`. Do that at the end of every sweep. It matters more than it sounds: the reference band is measured first, so it is the one most likely to be lost, and without it the rest of that instrument’s bands were measured against a different reference than the rest of your sample.

7 Every flag, explained

7.1 Shared by `resolve`, `catalogs`, `measure`, `spherex`, `run`

Flag	Takes	What it does	Default
<code>--name</code>	text	Resolvable object name; tried against Sesame, then NED, then SIMBAD. Use instead of <code>--ra/--dec</code> .	none
<code>--ra</code>	number	Right ascension in decimal degrees. Requires <code>--dec</code> .	none
<code>--dec</code>	number	Declination in decimal degrees. Requires <code>--ra</code> .	none
<code>--out-dir</code>	path	Directory everything lands under. Required on every verb except <code>resolve</code> and <code>remeasure</code> .	none
<code>--label</code>	text	Output filename stem.	sanitized name, or an IAU designation for a bare position

Give exactly one of `--name` or the `--ra/--dec` pair.

7.2 catalogs

Flag	Takes	What it does	Default
<code>--instruments</code>	one or more names	Archives to query: <code>allwise</code> , <code>galex</code> , <code>hst</code> , <code>jplus</code> , <code>legacy</code> , <code>panstarrs</code> , <code>sdss</code> .	none
<code>--all</code>	(switch)	Query every archive. Use this or <code>--instruments</code> .	off
<code>--radius</code>	arcsec	Starting cone-search radius. Doubles up to five times if nothing is found.	2.0
<code>--legacy-dr</code>	<code>dr9</code> or <code>dr10</code>	Legacy Surveys data release. <code>dr10</code> adds i-band in the south.	<code>dr9</code>
<code>--dered</code>	(switch)	Correct fluxes for Milky Way extinction. Off means as-measured, with the correction factor recorded per row.	off

7.3 measure

Flag	Takes	What it does	Default
<code>--instruments</code>	one or more names	Image archives: <code>cfht</code> , <code>hst</code> , <code>legacy</code> , <code>panstarrs</code> , <code>sdss</code> .	none
<code>--all</code>	(switch)	Every image archive. Use this or <code>--instruments</code> .	off
<code>--mode</code>	<code>aperture</code> or <code>sersic</code>	<code>aperture</code> reports the flux inside the circular aperture. <code>sersic</code> reports the fitted galaxy model's flux instead.	<code>aperture</code>
<code>--bands</code>	one or more filters	Filter subset applied to every archive.	each archive's own default set
<code>--aperture</code>	arcsec	Aperture radius. Must lie inside the <code>--radii</code> grid.	12.0
<code>--radii</code>	one or more arcsec	Radii at which the curve of growth is stored.	2 to 40 in 1-arcsec steps
<code>--cutout-size</code>	arcsec	Width of the image cutout. Half of this is the largest usable radius.	120
<code>--sky-rmin</code>	arcsec	Boundary between target and sky. Sky is measured only outside it. Lower it for compact targets.	15
<code>--registry</code>	path	JSON file of shared neighbor shapes to reuse (section 11).	none
<code>--registry-update</code>	(switch)	Also write this galaxy's results back into <code>--registry</code> . Requires <code>--registry</code> .	off
<code>--sersic-from</code>	band	With <code>--mode sersic</code> : fit the galaxy's shape on this band (<code>z</code> or <code>Legacy_z</code>) and use it for all bands.	per-archive refit

Flag	Takes	What it does	Default
<code>--sersic-params</code>	4 numbers	With <code>--mode sersic</code> : state the shape explicitly as n , axis ratio ($a/b \geq 1$), position angle (deg E of N), effective radius (arcsec).	none
<code>--sersic-seeing</code>	arcsec	PSF full width at half maximum assumed by a <code>--sersic-from</code> fit. The fitted index and radius are sensitive to it.	a typical value for that survey, not a measured one; the run prints a warning saying so
<code>--legacy-dr</code>	dr9 or dr10	Legacy Surveys release for both images and the scene catalog.	dr9
<code>--legacy-bricks</code>	(switch)	Download full Legacy brick coadds instead of small cutouts. Adds real per-pixel errors; roughly 40 MB per file.	off
<code>--hst-proposal-id</code>	text	Restrict HST to one observing program.	none
<code>--legacy-route</code>	auto, viewer, noirlab	Which service supplies Legacy cutouts. <code>auto</code> tries one and falls back to the other. See the warning below.	auto
<code>--dump-arrays</code>	(switch)	Write raw per-band arrays under <code><Instrument>/QA/</code> for debugging.	off

Pin the Legacy route for a sample you will compare

The two Legacy cutout services do not frame their images identically. Under `auto`, an outage at one of them silently sends some of your galaxies to the other, and the resampling shifts the measured star used for the PSF — which moves the fluxes by around a percent. That is invisible in the output. For any set of galaxies meant to be compared with one another, name the route explicitly; a pinned route reports an outage as an error instead of quietly substituting. Both the route you asked for and the route that actually answered are recorded for every band.

`--sersic-from`, `--sersic-params` and `--sersic-seeing` are refused under `--mode aperture`; the command tells you so rather than ignoring them.

7.4 spherex

Flag	Takes	What it does	Default
<code>--model</code>	psf or sersic	Source model for the forced photometry. <code>psf</code> biases extended sources.	sersic
<code>--sersic-params</code>	4 numbers	Explicit shape: n (≤ 6), axis ratio ($a/b \geq 1$), position angle (deg E of N), effective radius (arcsec).	Tractor catalog lookup
<code>--sersic-from</code>	band	Fit the shape on this band's image instead of the catalog lookup.	none

Flag	Takes	What it does	Default
<code>--sersic-seeing</code>	arcsec	PSF full width at half maximum of the shape-fit band.	a typical value for that survey, not a measured one
<code>--bkg-size</code>	pixels	Size of the tool's background-estimation region.	15
<code>--mjd-range</code>	2 numbers	Restrict to visits inside this MJD window. This is IRSA's documented workaround for epochs whose metadata breaks the job.	none
<code>--poll</code>	seconds	How often to ask IRSA whether the job is done.	5
<code>--timeout</code>	seconds	Give up on a job that is still <i>running</i> after this long.	10800 (3 h)
<code>--legacy-dr</code>	dr9 or dr10	Legacy release for the shape lookup or shape-fit image.	dr9

7.5 sed

Flag	Takes	What it does	Default
<code>--out-dir</code>	path	Directory holding the tables. Required.	none
<code>--label</code>	text	Which table family to plot.	inferred when the directory holds exactly one
<code>--spherex</code>	(switch)	Also draw the SPHEREx spectrum under the broadbands, as a <i>separate</i> figure named <code><label>_spherex_sed.png</code> . The broadband-only figure is left alone.	off

The SPHEREx figure draws the raw table exactly as IRSA returned it, with no quality cuts, which is deliberate: on a faint galaxy the red end visibly turns to noise, and seeing where that happens is useful. It is not what an SED fit would use.

Note that `run` does *not* write this figure — it is always a second, deliberate step. A galaxy with no `<label>_spherex_sed.png` has simply not had one made yet; it does not mean the galaxy has no SPHEREx data.

7.6 overlay

Flag	Takes	What it does	Default
<code>--out-dir</code>	path	Directory holding the tables. Required.	none
<code>--label</code>	text	Which table family to plot.	inferred when unambiguous

Flag	Takes	What it does	Default
<code>--zoom-size</code>	arcsec	Half-width of the zoom panel.	5
<code>--context-size</code>	arcsec	Half-width of the context panel.	15
<code>--wcs-from</code>	path	A local FITS file already on the composite's grid, used instead of downloading the ~380 MB detection mosaic. Refuses if the path names no file, or if the grid does not match.	none
<code>--dpi</code>	number	Figure resolution.	200

7.7 remeasure

The first argument is positional: the path to a `<label>_measured.provenance.json`.

Flag	Takes	What it does	Default
<code>--mode</code>	sersic or aperture	sersic reports the fitted model's flux; aperture reports the measured, neighbor-subtracted flux. Matches the measure default, so a bare remeasure re-reports the same quantity the measurement produced.	aperture
<code>--aperture</code>	arcsec	Radius to report at. Matches the measure default, so a bare remeasure re-reports at the radius the measurement used.	12
<code>--integrated</code>	(switch)	Report the total instead of an aperture. In sersic mode that is the model's total; in aperture mode it is the outermost stored radius. Overrides <code>--aperture</code> .	off
<code>--shape</code>	forced or fitted	Which galaxy shape the report is built on. forced uses the archive's reference-band shape. fitted uses each band's own shape where one was stored; bands without one fall back to forced and say so.	forced
<code>--registry</code>	path	Registry file to fall back on when rebuilding past the stored radius grid. Unlike measure , remeasure finds one on its own when you do not name it (section 11.2).	a registry.json beside the galaxy directory, if present
<code>--write-qa</code>	(switch)	When rebuilding past the grid, write per-band figures to <code>QA/remeasure_R<N>as/</code> .	off
<code>--out</code>	path	Write the result to this CSV instead of printing it.	print

7.8 run

run takes every flag from **catalogs** (`--radius`, `--dered`, `--legacy-dr`) and every flag from **measure** **except** `--instruments`, `--all` and `--dump-arrays`, which it replaces with its own archive selection, plus:

Flag	Takes	What it does	Default
<code>--catalogs</code>	one or more names, or all / none	Which catalog archives to query. none skips the catalog stage entirely.	all
<code>--images</code>	one or more names, or all / none	Which survey images to fetch and measure. none skips the measurement stage entirely.	all
<code>--skip</code>	one or more names	Archives to leave out. Accepts both catalog and image archive names, and applies to both stages.	none
<code>--spherex</code>	off, psf, sersic	Also fetch SPHEREx, with this source model.	off
<code>--spherex-bkg-size</code>	pixels	Background region for the SPHEREx job.	15
<code>--spherex-mjd-range</code>	no numbers	Restrict the SPHEREx job to visits in this MJD window.	none
<code>--spherex-timeout</code>	seconds	Give up on a running SPHEREx job after this long.	10800

run selects archives with `--catalogs` and `--images` rather than **measure**'s `--instruments`, because it drives two stages that draw on different archives.

Name the image archives for any real campaign

`--images` defaults to *all*, and *hst* is in that set. A bare **run** therefore queries the Hubble archive for every target, which downloads hundreds of megabytes per galaxy and, for a typical ground-based target, measures nothing at all. Name what you want: `--images legacy cfht sdss panstarrs`.

One special case: `--sersic-params` is normally refused under `--mode aperture`, but under **run** with `--spherex psf` or `--spherex sersic` it is accepted, because there it also declares the shape SPHEREx extracts with. One shape per galaxy, shared by both stages.

7.9 plan

Flag	Takes	What it does	Default
<code>--targets</code>	path	CSV of targets. Required.	none
<code>--out</code>	path	Plan JSON to write; a <code>.csv</code> of the same rows lands beside it. Required.	none
<code>--out-root</code>	path	Root that the CSV's <code>dir</code> column is taken relative to.	the current directory
<code>--cutout-size</code>	arcsec	Stamp width the plan is computed for. The sweep must use the same one.	120

Flag	Takes	What it does	Default
<code>--link-margin</code>	arcsec	Extra separation required before two galaxies are treated as independent. Larger is safer and slower.	120

7.10 batch

Takes every measurement flag `measure` takes, minus the registry flags (the plan decides those), plus:

Flag	Takes	What it does	Default
<code>--plan</code>	path	Plan from <code>plan</code> . Required.	none
<code>--registry-dir</code>	path	Where the per-group registries and the merged one are written. Required.	none
<code>--report</code>	path	Per-target report CSV. Required.	none
<code>--workers</code>	number	Galaxies measured at once.	4
<code>--pass</code>	harvest, parallel, all	Run only one of the two passes.	all
<code>--no-groups</code>	(switch)	Run the harvest pass as one sequence instead of in parallel groups. Slower; the reference the grouped version is checked against.	off
<code>--no-resume</code>	(switch)	Re-measure galaxies that already have products.	off
<code>--log-dir</code>	path	Per-target logs land here, written line by line.	none
<code>--stop-after-fail</code>	number	Abandon the sweep after this many galaxies fail. 0 disables.	20

7.11 spherex-plan

Flag	Takes	What it does	Default
<code>--targets</code>	path	CSV of targets — the same file <code>plan</code> reads. Required.	none
<code>--out</code>	path	Config JSON to write; a <code>.csv</code> of every member lands beside it. Required.	none
<code>--out-root</code>	path	Root the CSV's <code>dir</code> column is relative to.	the current directory
<code>--data-root</code>	path	Root recorded <i>in</i> the config, relative to the config file.	.
<code>--blend-radius</code>	arcsec	Targets closer than this share a job. Grouping chains: A near B near C puts all three together.	45
<code>--model</code>	psf or sersic	Source model for every job. One model per job — the tool has no per-source switch.	sersic

Flag	Takes	What it does	Default
<code>--no-companions</code>	(switch)	Group targets only; do not add catalog neighbors.	off
<code>--companion-flux-ratio</code>	number	Add a neighbor when it is at least this fraction of the target's brightness.	0.1
<code>--bkg-size</code>	pixels	Background region recorded in the config.	15
<code>--mjd-range</code>	two numbers	Visit window recorded in the config.	none
<code>--cutout-size</code>	arcsec	Stamp width the scene catalogs were cached at.	120
<code>--legacy-dr</code>	dr9 or dr10	Catalog release the shapes come from.	dr9
<code>--description</code>	text	Free text recorded in the config.	none

7.12 `spherex-batch`

Flag	Takes	What it does	Default
<code>--config</code>	path	Config from <code>spherex-plan</code> . Required.	none
<code>--report</code>	path	Per-group report CSV. Required.	none
<code>--workers</code>	number	Jobs submitted at once. The service runs two at a time and queues the rest rather than refusing them, so a larger number is allowed — the extras simply wait, and waiting does not count against a job's run time.	1
<code>--only</code>	one or more ids	Run just these groups, e.g. after editing them.	all
<code>--no-resume</code>	(switch)	Re-enter groups already on disk. Existing tables are still never overwritten.	off
<code>--log-dir</code>	path	Per-group logs land here, written line by line.	none

8 What comes out

8.1 The directory tree

A full run of one galaxy produces:

```

<out-dir>/
+-- patches.json                (optional, you create it)
+-- Photometry/
    +-- <label>_catalog.csv      archive photometry
    +-- <label>_catalog.provenance.json
    +-- <label>_measured.csv     measured photometry
    +-- <label>_measured.provenance.json
    +-- <label>_sed.png          combined SED figure
    +-- <label>_spherex_sed.png  with SPHEREx (sed

```

```

--spherex)
  +-- <label>_overlay.png           position check figure
  +-- coverage_catalogs.json       per-archive outcome
  +-- coverage_measure.json        per-archive outcome
  +-- Legacy/      cached images + QA/Legacy_<band>.png
  +-- SDSS/        cached images + QA/SDSS_<band>.png
  +-- PanSTARRS/   cached images + QA/PS1_<band>.png
  +-- CFHT/        cached images + QA/CFHT_<band>.png
  +-- HST/         cached mosaics + QA/HST_<band>.png
  +-- QA/
  |   +-- growth_curves.png        all bands, one figure
  +-- scene/                      cached catalog queries
  +-- SPHEREx/
      +-- table_photometry.<tag>.csv
      +-- table_photometry.<tag>.provenance.json
      +-- extractions.json

```

The `<tag>` on a SPHEREx table names the exact configuration it was extracted with. A table from a joint fit (section 6.15) is tagged `joint-<model>-<hash>` instead of `<model>-<hash>`, so the two sit side by side in the same directory and nothing can confuse one for the other.

The list verbs write outside any one galaxy:

```

plan.json, plan.csv           from plan
groups.json, groups.csv      from spherex-plan
report.csv                   one row per galaxy or group
logs/<name>.log               one per galaxy or group
registries/                  from batch
  +-- group_<n>.json          one per harvest group
  +-- registry.json           the merged, frozen copy
spherex_groups/              from spherex-batch
  +-- joint_photometry.<tag>.csv the whole joint table,
                                and any ancillary spectra

```

8.2 What each file is

File	What it is
<code><label>_catalog.csv</code>	One row per band, from the published archive catalogs. The nearest source to the requested position in each archive.
<code><label>_measured.csv</code>	One row per band, measured by <code>sedphot</code> from the images. Same column layout as the catalog table.
<code>*.provenance.json</code>	The complete record of how the neighboring file was made: every setting, the code revision, a checksum of the file, and for measurements the full fit. This is what <code>remeasure</code> reads. Do not edit it.
<code><label>_sed.png</code>	Flux against wavelength, both tables together. Catalog points are filled, measured points are open. Marker shape identifies the instrument, color tracks wavelength.

File	What it is
<label>_overlay.png	Two panels of the HST color image with each archive's matched position marked.
coverage_catalogs.json, coverage_measure.json	Per-archive outcome: <code>ok</code> , <code>no_coverage</code> , <code>no_match</code> , or <code>error</code> , with a message and the number of rows.
<Instrument>/*.fits	The downloaded images, cached. Deleting one forces a re-download on the next run.
<Instrument>/QA/ <Inst>_<band>.png	The per-band diagnostic figure (section 9). <Inst> is the instrument's short name, which matches the directory everywhere except PanSTARRS/, whose figures are named PS1_<band>.png.
QA/growth_curves.png	Enclosed flux against radius for every measured band on one axis.
scene/*.csv	Cached catalog queries describing the neighboring sources. Deleting these forces a re-query.
SPHEREx/table_photometry. <tag>.csv	The raw per-visit SPHEREx table, exactly as IRSA returned it. Never overwritten.
SPHEREx/extractions.json	An index mapping each tag to the configuration that produced it.

8.3 The output table columns

Both <label>_catalog.csv and <label>_measured.csv share this layout.

Column	Meaning	Units
band	Band label, <Inst>_<filter> — e.g. Legacy_g, SDSS_u, PS1_z, CFHT_r, GALEX_FUV, WISE_W1, JPLUS_uJAVA, HST_F475W.	—
flux_uJy	The flux.	microjansky (AB)
flux_err_uJy	Uncertainty on the flux.	microjansky
mag_AB	The same measurement as an AB magnitude. Empty when the flux is zero or negative.	mag
mag_err	Uncertainty on the magnitude.	mag
target_ra	The position you asked for.	degrees
target_dec	The position you asked for.	degrees
match_ra	Where the archive's matched source actually is. For measured rows this equals <code>target_ra</code> .	degrees
match_dec	Where the archive's matched source actually is.	degrees
sep_arcsec	Distance from the requested position to the match. 0 for measured rows.	arcsec
flags	Quality tokens, <code>key=value</code> separated by semicolons (see below). For catalog rows, whatever flags that archive publishes.	—
source	Where the number came from — Legacy_DR9, SDSS_DR17_cModel, PanSTARRS_DR1, GALEX_GUVcat_AIS, sedphot_aperture_scene_empap, and so on.	—
retrieved	Date the value was pulled.	ISO date

Column	Meaning	Units
<code>mw_transmission</code>	Milky Way transmission in this band, where the archive supplies it. Empty when unknown.	fraction
<code>dered_applied</code>	True if the flux has been corrected for Milky Way extinction. Measured rows are always False.	—

Three things to know about the numbers.

- **Fluxes are microjansky on the AB system.** Convert to AB magnitude with $\text{mag} = 23.9 - 2.5 \log_{10}(\text{flux})$.
- **The errors are statistical only.** They describe the noise in the measurement and nothing else — no calibration uncertainty, no error floor, no model uncertainty. Adding those is the SED fitter’s job, not this table’s.
- **Negative fluxes are real.** A non-detection in a faint band gives a small negative number after background subtraction. That is the honest measurement and it is preserved deliberately; it is not an error, and it must not be clipped to zero. Its `mag_AB` cell is simply empty.

The flags tokens on measured rows. These let you filter a table without opening a single figure.

Token	Meaning	What to look for
<code>cov</code>	Fraction of aperture pixels with real data.	Should be 1.000. Below 0.95 the band is refused outright.
<code>maskfrac</code>	Fraction of the aperture masked because a neighboring source sat there.	A few percent is normal. Above ~0.2 the flux leans heavily on reconstruction.
<code>twinfrac</code>	Fraction of masked pixels reconstructed from the galaxy’s mirror side rather than from the model.	1.00 is best.
<code>nbsub</code>	Flux subtracted from the aperture as neighboring sources.	Compare against the reported flux. Large values mean a crowded field.
<code>excess</code>	Flux growing outside the aperture beyond what the galaxy’s own model predicts.	Near zero is good. Large positive means contamination or an unmodeled envelope.
<code>pedb</code>	Residual uniform background left in the curve of growth.	Should be near zero.
<code>conv</code>	Radius, in arcsec, at which the curve of growth flattened. 0 means it never did.	Should be at or inside the aperture.
<code>bg</code>	The fitted background level.	—
<code>far</code>	An independent background level measured far from the target.	Should broadly agree with <code>bg</code> .
<code>mesh</code>	Flux the smooth background surface removed from the aperture.	Large values mean structured background.

Token	Meaning	What to look for
art	Area, in square arcsec, flagged as a detector artifact (a bleed trail, a streak).	Present only when one was found.
leash	How many neighboring sources hit their brightness limit.	Present only when some did.
refit	The galaxy's fitted brightness relative to the catalog's.	Near 1 is unremarkable.
atbound	How many fitted shape parameters ended at a limit.	Present only when some did.
reg	How many neighbors came from the shared registry.	—
scene=none	No catalog of neighboring sources was available; the galaxy was measured with no neighbor modeling at all.	Treat the flux with care in a crowded field.

9 Reading the QA figures

Two figures per measurement run, plus one per band.

9.1 The per-band figure

`<Instrument>/QA/<Inst>_<band>.png` — five panels in a row. The prefix is the instrument's short name, which matches the directory everywhere except `PanSTARRS/`, whose figures are `PS1_<band>.png`.

1. **data** — the image, with stars already subtracted. The blue circle is your aperture. Orange contours, if present, mark detector artifacts. The title gives the fitted background level and its tilt.
2. **fitted scene + background** — the model: the galaxy, its neighbors, and the background, as the fit believes them to be.
3. **residual - mesh** — the data minus the model, and minus the smooth background surface. This is the panel that tells you whether the fit worked.
4. **masked + filled - mesh** — the image the flux is actually measured from: masked pixels replaced by reconstructed values. Green contours outline the mask. The title gives the masked fraction of the aperture.
5. **curve of growth** — enclosed flux against radius. The red line with dots is the measurement; the black dashed line is the fitted galaxy model. The blue vertical line is your aperture; a grey dotted line marks where the curve flattened. The title gives the final flux, its error, and the excess growth.

A good figure.

- Panel 3 is flat noise. No bright leftover at the galaxy's center, no dark hole where a neighbor was over-subtracted.
- Panel 5's red curve rises and then flattens, and the flattening happens at or inside the blue aperture line. The title says `conv@14"` or similar rather than `no plateau`.
- The red curve and the black dashed model track each other out to the aperture.

- **excess** in the title is small compared with the flux.
- Green mask contours in panel 4 cover only a small part of the blue circle.

A suspicious figure.

What you see	What it means
Title says no plateau	The curve of growth was still rising at 40 arcsec. Either the galaxy is genuinely large, or the background is under-subtracted. Check pedb and the far token.
Red curve turns over and falls	The background was over-subtracted.
Large positive excess	Light in the outskirts that the galaxy's model does not account for — a neighbor, or a real envelope.
Dark hole at a neighbor's position in panel 3	That neighbor was over-subtracted.
Bright residual core in panel 3	The galaxy profile did not fit; the flux may be biased.
Green contours across much of the blue circle	Most of the aperture is reconstructed rather than measured. Check twinfrac .
Orange artifact contours crossing the aperture	A detector artifact sits on the target.
Red and black curves diverge inside the aperture	The fitted model is a poor description of the galaxy. Fine in --mode aperture (the model is only a cross-check there), serious in --mode sersic (the model <i>is</i> the answer).

9.2 The growth-curve figure

QA/growth_curves.png — every measured band on one axis, enclosed flux against radius, with a logarithmic vertical scale. Line style identifies the instrument, color tracks wavelength. The blue vertical line is the science aperture.

Read it for consistency across bands. Curves should be roughly parallel and should flatten at similar radii. A single band that keeps climbing while its neighbors flatten, or one that crosses the others, is the band to open individually.

10 When something goes wrong

Every command reports failures rather than hiding them. Most messages name the fix. The recurring ones:

Symptom	Cause	Fix
error: the following arguments are required: --out-dir	--out-dir is required on the six verbs that write into one galaxy's directory.	Add --out-dir <directory>. The list verbs name their outputs differently — see section 4.
error: give --instruments ... or --all	catalogs and measure need to be told which archives to use.	Add --all, or --instruments legacy sdss.
Could not resolve 'X' with Sesame, NED, or SIMBAD	The name is not in any resolver.	Look the position up yourself and use --ra / --dec.

Symptom	Cause	Fix
The run finished, but a galaxy has fewer bands than I asked for	A band failed on its own — a dropped download, a slow disk — and the run logged it and carried on. The galaxy still reports ok.	Count the rows in <code><label>_measured.csv</code> . Re-run that galaxy; cached images make it quick. This is why you check band counts rather than the report.
A joint SPHEREx group failed with a shape disagreement	The tool reported using a different shape from the one asked for, so every member's flux is against an unknown model.	Nothing was written. The raw table is kept beside the group as <code>...unverified.csv</code> so the hour of computing is not lost. Check the shape columns in your <code>groups.csv</code> and run the group again with <code>--only</code> .
<code>--registry-update</code> needs <code>--registry PATH</code>	You asked to write to a registry without naming one.	Add <code>--registry <path></code> , or drop <code>--registry-update</code> .
<code>--sersic-from</code> only applies to <code>--mode sersic</code>	A shape flag was given under aperture mode, where it would have no effect.	Add <code>--mode sersic</code> , or drop the flag.
<code>ValueError: --aperture 60" lies outside the curve-of-growth grid [2, 40]"</code>	The aperture is outside the radius grid the run stores.	Pass <code>--radii</code> covering it, and a <code>--cutout-size</code> at least twice the largest radius.
An image archive reports <code>no_coverage</code>	That survey's footprint does not include this position. Nothing is wrong.	Nothing to do. Fewer bands is the correct answer.
A catalog archive reports <code>no_match</code>	Either the field genuinely holds no catalogued source within the expanded search radius (including "this position is outside that survey's footprint"), or the archive was down. The two are indistinguishable: a failed query returns no rows, and the radius expansion then reports <code>no_match</code> . Catalog archives never report <code>no_coverage</code> .	Re-run just that archive later. If it still says <code>no_match</code> , try a larger <code>--radius</code> . An unexpected <code>no_match</code> on a well-covered galaxy is almost always an outage.
An archive reports error	A service or parsing failure that got past the provider's own retries. The message names it.	Read <code>coverage_catalogs.json</code> or <code>coverage_measure.json</code> , then re-run that archive.

Symptom	Cause	Fix
Legacy z: no_coverage -- aperture coverage 0.83 < 0.95	More than 5% of the aperture is off the image footprint, blank, or covered by an artifact. The band is refused rather than reporting a silently biased flux.	Use a smaller <code>--aperture</code> , or a larger <code>--cutout-size</code> if the cutout was simply too small. Otherwise accept the missing band.
blank pixels inside the 0.5" peak	Dead pixels at the very center of the galaxy. No reconstruction can invent the peak.	The band cannot be measured.
scene catalog is empty here; measuring blind, and scene=none in flags	No catalog of neighboring sources covers this position, so nothing was subtracted or masked.	The flux is still measured, but with no neighbor handling. In an uncrowded field this is fine; in a crowded one, treat it with care.
No images fetched from any provider; nothing to measure.	No archive covered the position, or all of them failed.	Check <code>coverage_measure.json</code> for which and why.
overlay downloads ~380 MB	The HST color composite carries no coordinate grid of its own, so the detection mosaic is fetched for its grid. It is cached after the first time.	Pass <code>--wcs-from</code> with a local FITS file already on that grid.
no HAP total/detection product at this position, or sedphot overlay: no figure written <code>--wcs-from</code> names no file	HST has not imaged this field.	The overlay is unavailable for this galaxy.
[spherex] Tractor shape lookup failed ... aborting or no usable extended Tractor shape at this position table_photometry. <tag>.csv exists but no sidecar vouches for this configuration	The path is mistyped. It refuses rather than falling back to the big download. SPHEREx will not guess a source shape. A raw table written under the wrong shape is permanent and unmarked, so it stops instead. A raw SPHEREx table is on disk that nothing certifies. Nothing is ever overwritten.	Correct the path. Pass <code>--sersic-params N AXR PA RE</code> , or <code>--sersic-from z</code> to fit one, or <code>--model psf</code> . Move the file aside deliberately, then re-run.

Symptom	Cause	Fix
[spherex] WARNING: reff 0.60" is below the tool's 1" point-source threshold	The tool treats anything that small as a point source, so the Sérsic parameters do nothing.	Either accept it, or use <code>--model psf</code> and be explicit.
remeasure prints aperture 45" is past the stored curve of growth (ends at 40"); rebuilding the scene	Beyond the stored grid there is no value to interpolate, so the scene is rebuilt and re-integrated.	Nothing wrong — just expect minutes rather than milliseconds.
sedphot remeasure: aperture 70" exceeds the stamp half-width (60")	The requested radius runs off the cutout the fit was built on.	Re-measure with a larger <code>--cutout-size</code> .
remeasure says bands fell back to the forced shape under <code>--shape fitted</code>	Those bands have no per-band shape stored — either an older measurement, or a galaxy where the extra fit never ran.	Either accept <code>forced</code> , or re-measure the galaxy.
cannot infer <code>--label</code> in ...: found stems [...]	The directory holds tables for more than one galaxy, so <code>sed</code> and <code>overlay</code> cannot guess which.	Pass <code>--label <stem></code> .
[sed] WARNING ... the plotted fluxes are NOT on a common extinction scale	You used <code>--dered</code> , which corrects the catalog table only. The measured table is always as-measured, so the figure mixes the two.	Expected. Do not read colors off that figure; use the tables.

11 Getting more out of it

Two features exist for cases the defaults do not cover. You will not need either at first.

11.1 patches.json — correcting the scene by hand

Before measuring, `sedphot` builds a picture of the field from the Legacy Surveys catalog: which sources are there, how big and how bright. Occasionally that catalog is wrong in a way that matters — most often it merges a close pair into a single entry, or splits one galaxy into several.

You can correct it. Put a file named `patches.json` in the galaxy's `--out-dir` (next to `Photometry/`, not inside it). It is read automatically and its keys are logged at the start of the run.

The keys it accepts:

Key	What it does
<code>replace_rows</code>	Replace one catalog entry with one or more of your own. The usual fix for a merged close pair.
<code>free_seats</code>	Let the source at a given position have its shape fitted rather than taken from the catalog.
<code>target_system</code>	Declare that a nearby source belongs to the target — its light is counted with the galaxy rather than subtracted from it.
<code>target_halo</code>	Give the target an extended-envelope component, for a brightest-cluster galaxy.
<code>target_refit</code>	Turn the target's own shape refit on or off.
<code>snap_gated</code>	Move fitted sources onto their nearest brightness peak.
<code>harvest_target</code>	Record the target's own shape into the shared registry.
<code>comment</code>	Free text. Ignored by the code; use it to say why the patch exists.

Reach for this when a QA figure shows a clear over- or under-subtraction at a specific neighbor, and only then. An unnecessary patch is a way to bias a measurement.

11.2 The cross-field registry — shared neighbors

When several galaxies in the same field are measured separately, a neighboring source may appear in more than one of them. Left alone, each measurement fits that source independently, and the fits disagree.

The registry fixes that. It is one JSON file recording the fitted shape, position and per-band brightness of shared sources, in sky units so an entry works from any field.

Consume it:

```
sedphot measure --name "M87" --all --out-dir M87 \
  --registry registry.json
```

Consume it and write this galaxy's results back into it:

```
sedphot measure --name "M87" --all --out-dir M87 \
  --registry registry.json --registry-update
```

Three things to know before using it:

- **Driving measure yourself, run galaxies one at a time.** Updates are last-writer-wins with no file locking, so two runs writing at once will corrupt the file. This is a limitation of the flag, not of the package: `batch` (section 6.17) measures many galaxies at once safely, by giving each group of galaxies its own registry file and merging them afterwards.
- **Order matters while the file fills.** A galaxy measured after its neighbor was registered gets that neighbor's shape frozen in rather than fitted fresh. Once every galaxy in a field has been measured once, a second pass over the same file is uniform.
- **remeasure finds a registry on its own; every other verb waits to be told.** `measure` uses a registry only when you pass `--registry`, because consuming one changes the fluxes and a fresh measurement should not do that unasked. `remeasure` is the exception: when it has to rebuild the scene past the stored radius grid, it picks up a `registry.json` sitting beside the galaxy directory without being asked, so that a rebuild reuses the shapes the original measurement was built against. It prints the file it found; pass `--registry` to name a different one.

Use it when you are measuring several galaxies in one field and want their shared neighbors treated identically. Skip it for a single galaxy; it buys nothing there.

Note

Every number this package writes is accompanied by a `.provenance.json` file recording the settings, the code revision, and a checksum of the product. When a result needs defending, that file is the record.